“Cricket Player Performance Prediction”

Presented By

Soumya S Betgeri

Abstract

Cricket analytics is growing fast with more match data and better computing. This project builds a simple model to predict a player's runs and wickets using past stats, recent form, venue info, and opponent details. A clear dashboard shows predictions and trends, turning raw data into helpful insights.

Problem Statement

Cricket performance changes with many moving parts—from form and conditions to opponent strength. This project builds a smart predictive system that studies past matches to estimate runs, wickets, and key factors, helping teams and fans make better choices.

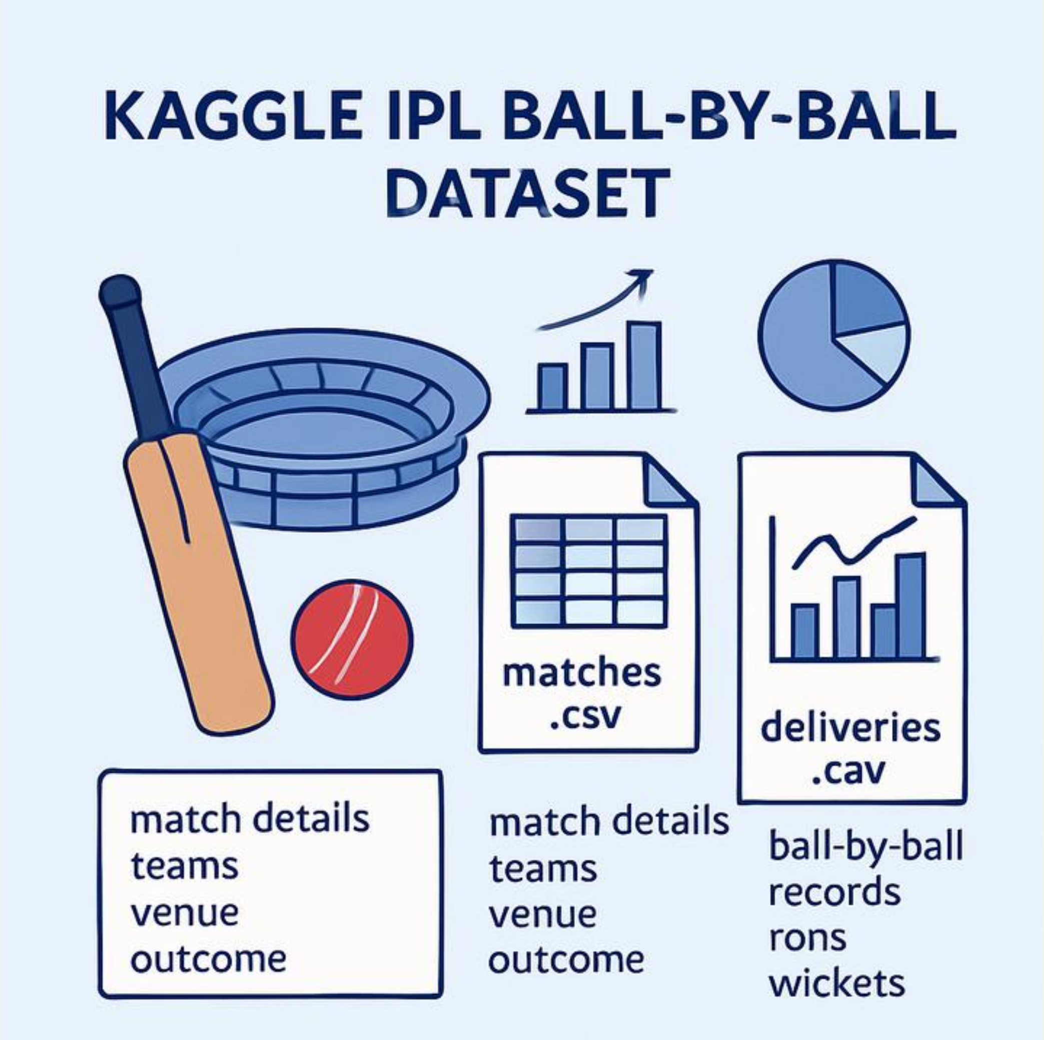
Objectives

- Collect and preprocess historical cricket data
- Identify key performance indicators affecting player output
- Develop machine learning models for prediction
- Evaluate prediction accuracy using statistical metrics
- Build an interactive dashboard for visualization
- Provide explainability using feature importance

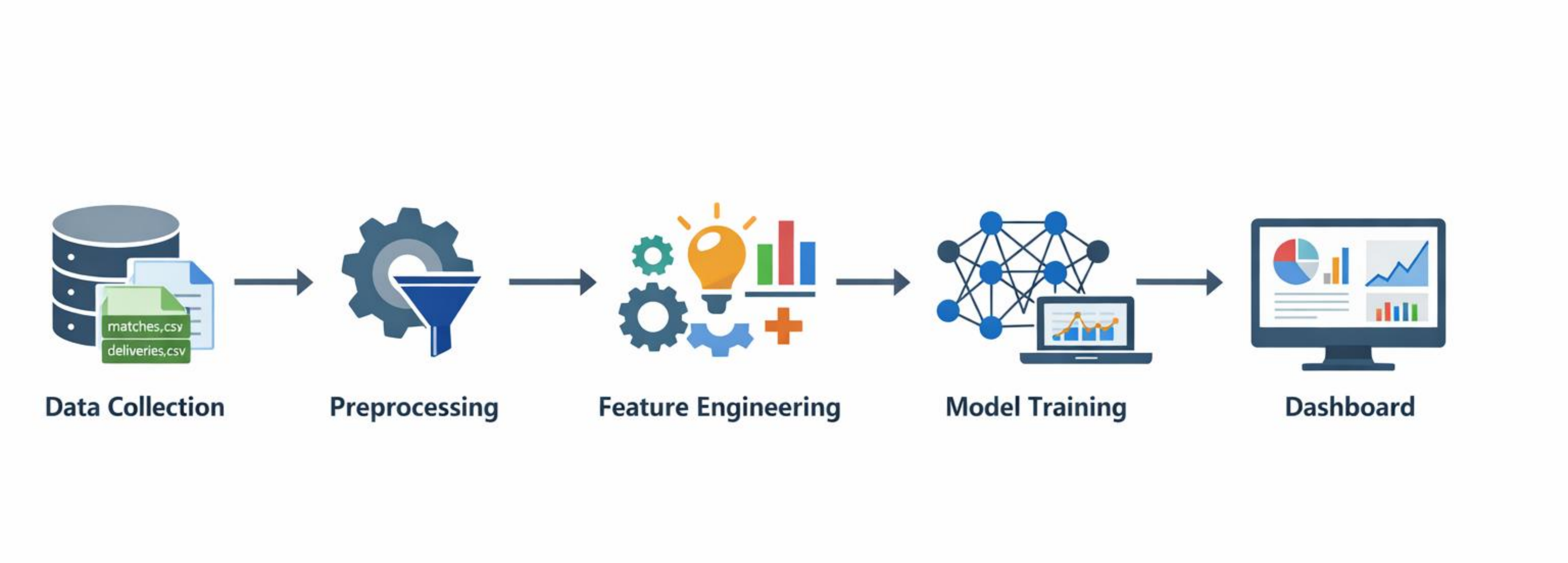
Dataset Description

Source: Kaggle IPL Ball-by-Ball Dataset

matches.csv
Size: 1,095 rows × 20 columns
Details: match ID, season, date, venue, teams, toss details, and match outcome
deliveries.csv
Size: 260,920 rows × 17 columns
Description: Contains detailed ball-by-ball records including batting team, bowling team, runs scored, extras, and wicket events



System Architecture

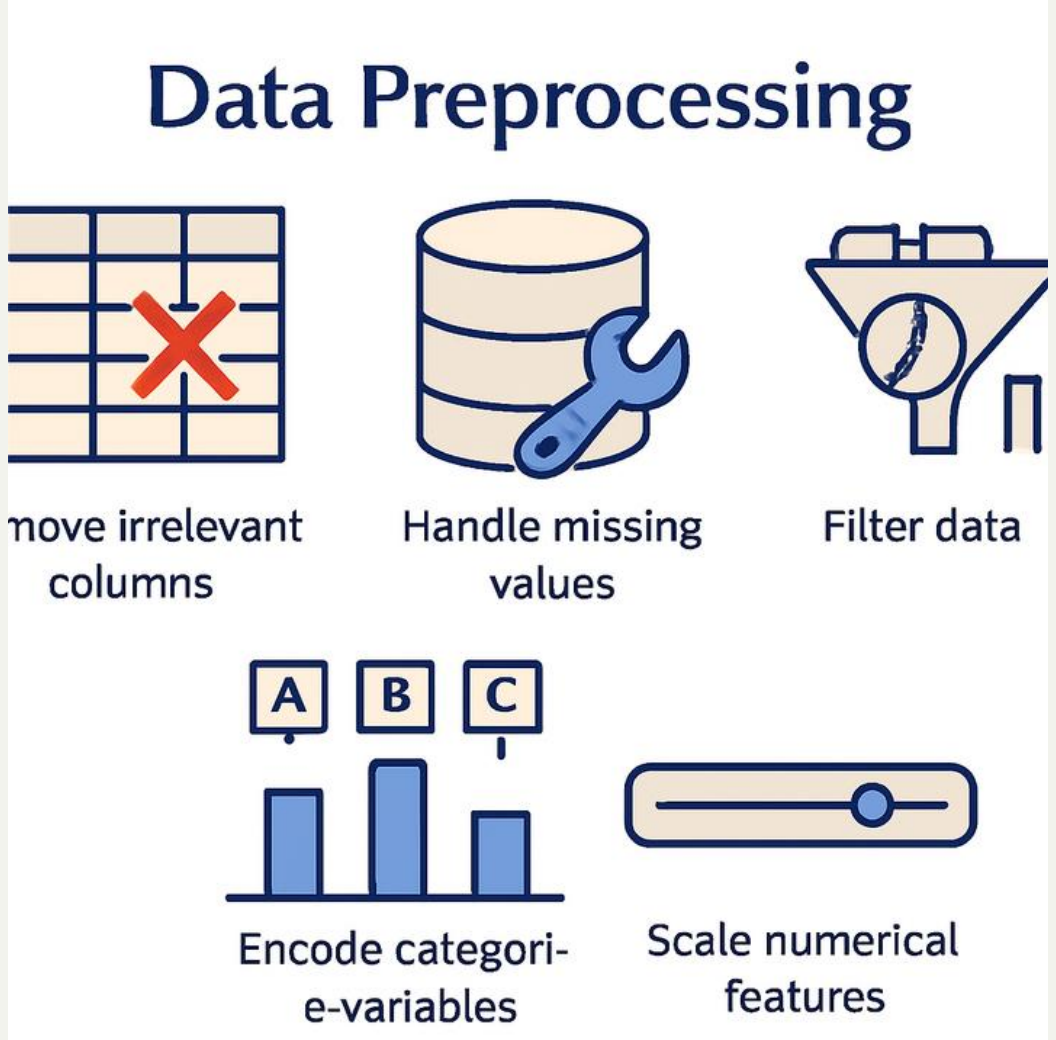


This pipeline ensures systematic data flow from raw dataset to final prediction.

Methodology

1.Data Collection:

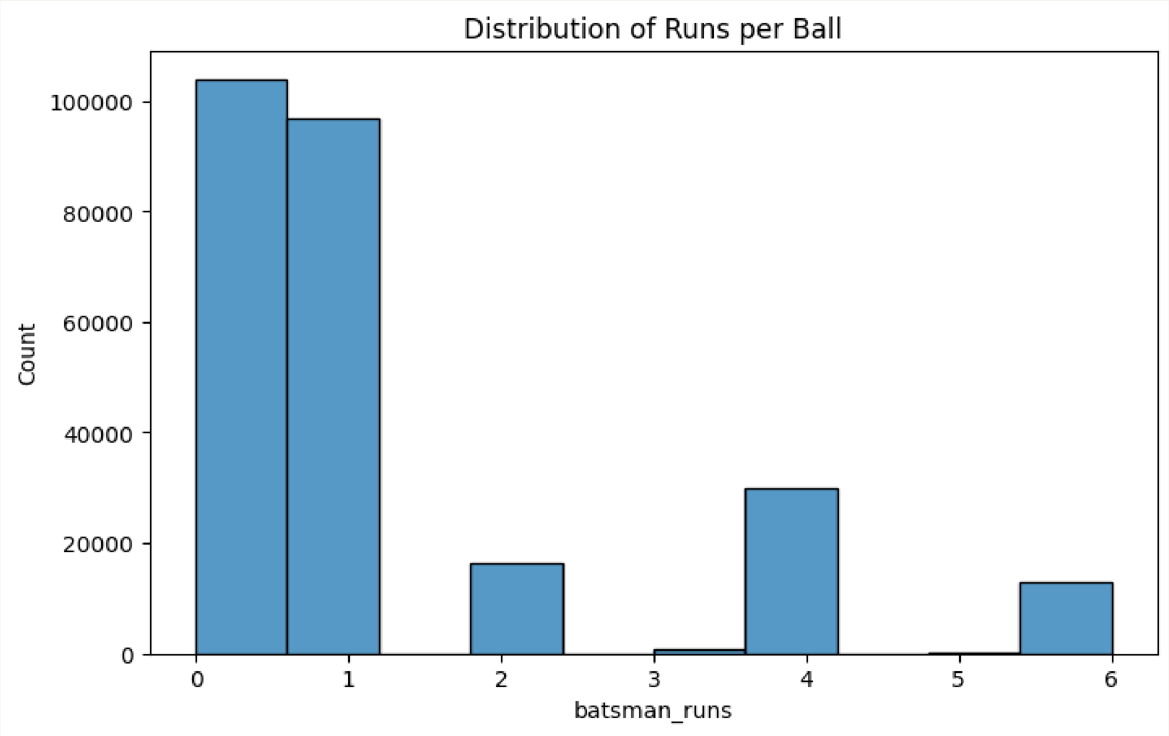
- Historical cricket datasets were obtained containing ball-by-ball and match-level statistics.
- The datasets include multiple seasons to ensure sufficient training samples and variability



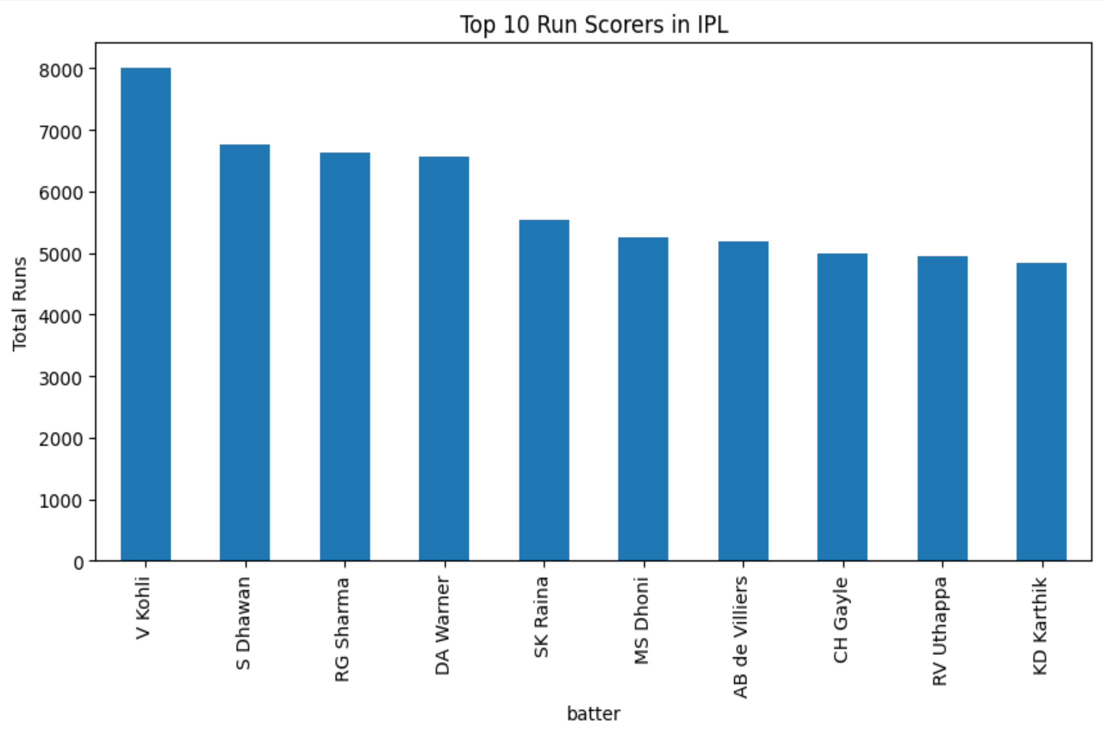
2.Data Preprocessing:

- Removed irrelevant columns
- Handled missing values
- Filtered players with low match counts
- Scaled numerical features

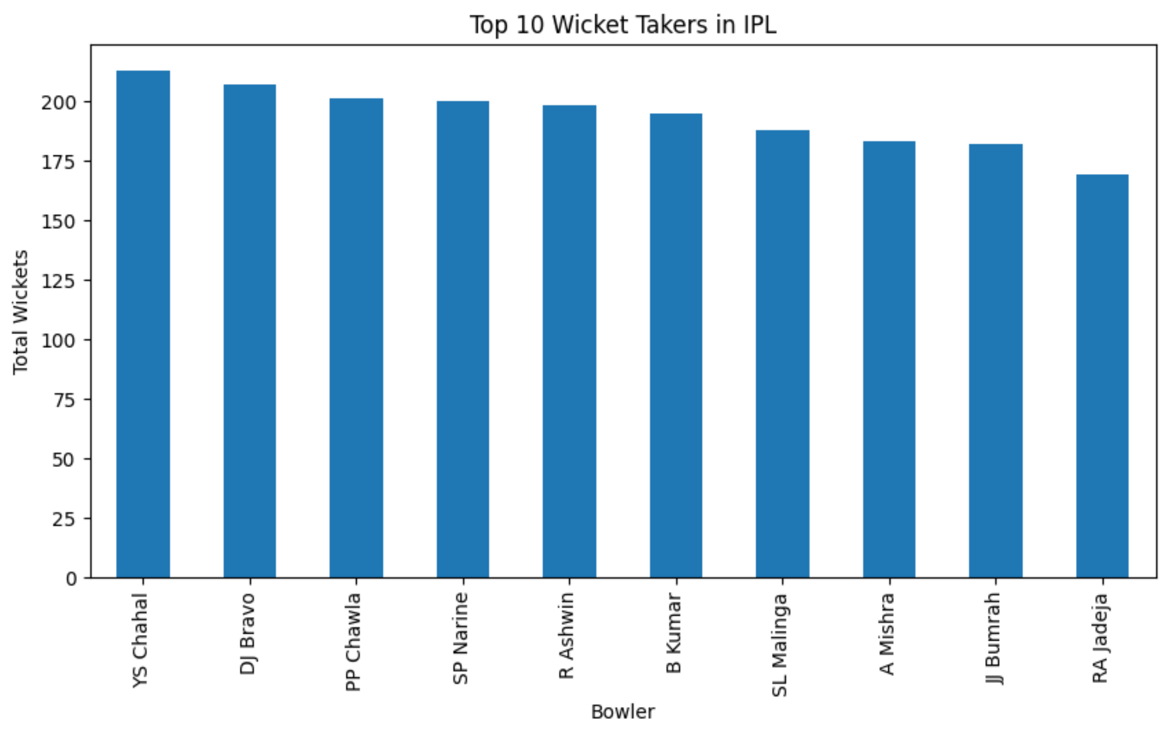
Exploratory data analysis



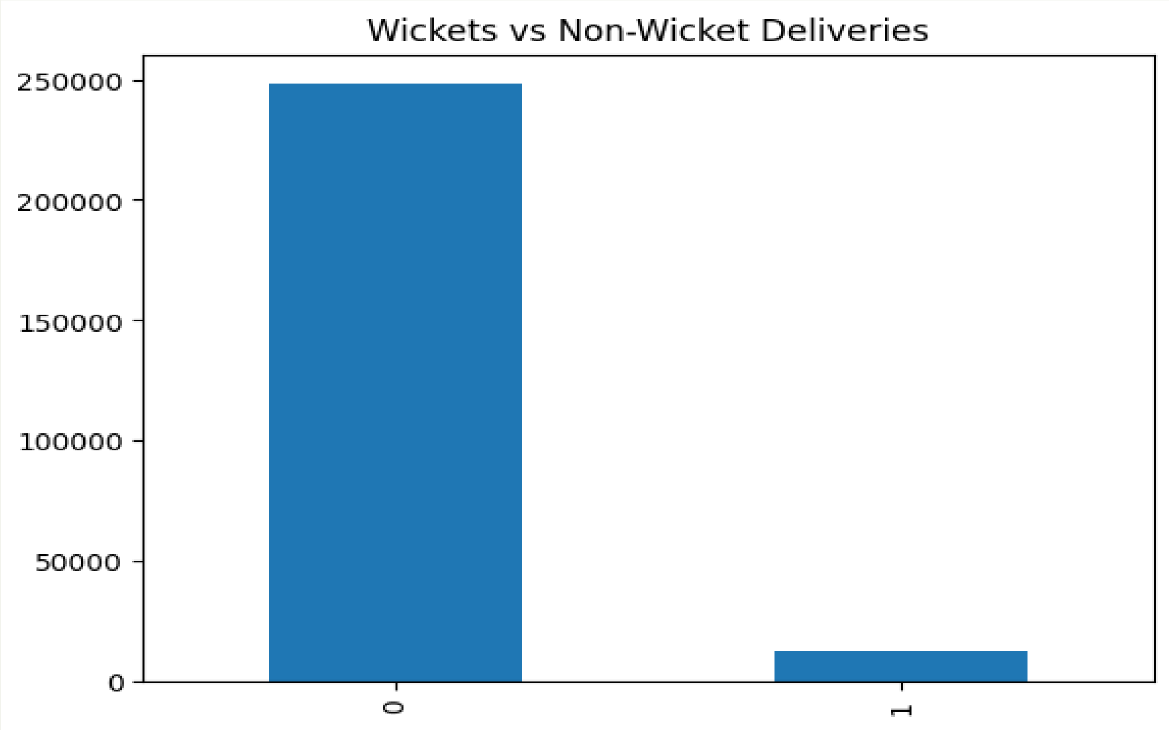
EDA – Runs Distribution



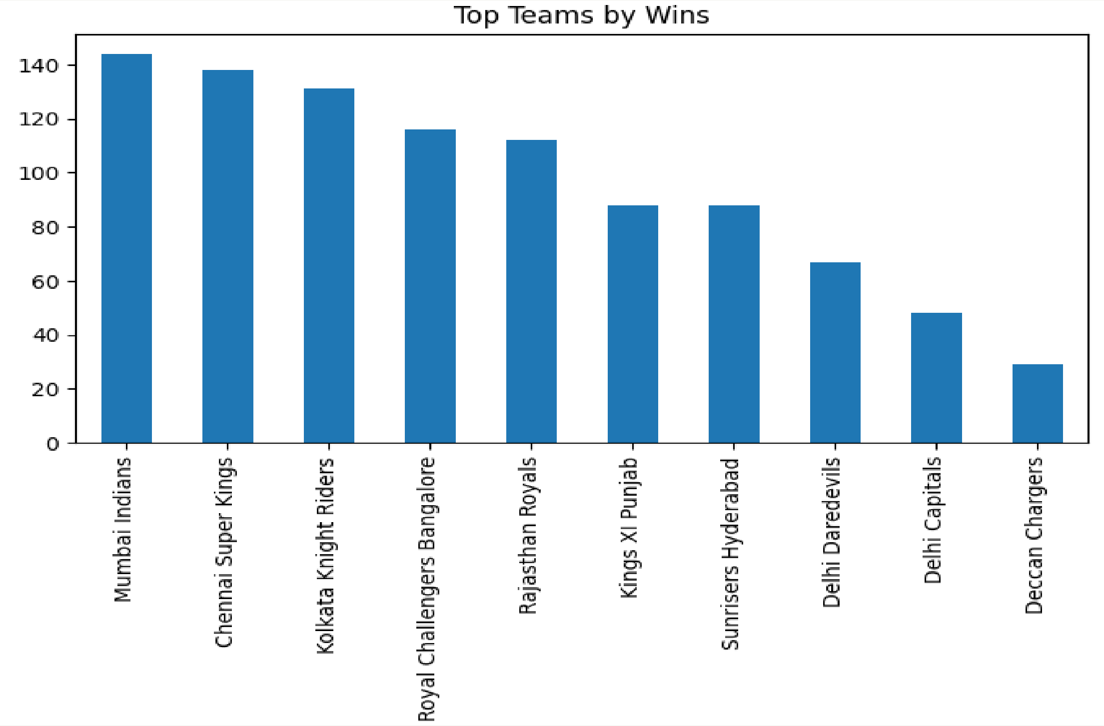
EDA- Top 10 Run Scorers in IPL



EDA- Top 10 wicket takers in IPL



EDA- Wicket Distribution



EDA-Team Performance

3.Feature engineering

Career average- runs represents long-term ability

Recent form average- performance in last few matches

Venue average - performance at specific stadiums

Opponent average - performance against specific teams

Run consistency- variability in scoring

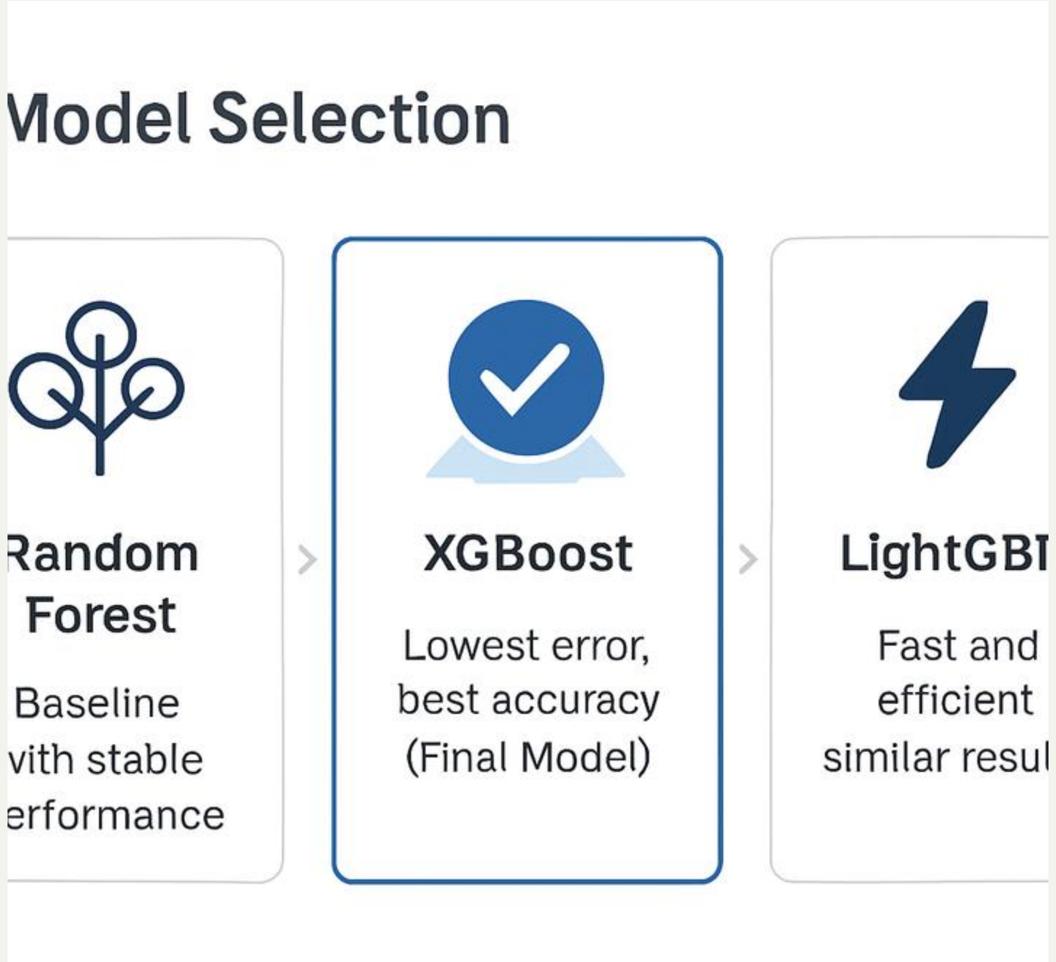
Matches played - experience indicator

Wicket average- bowling effectiveness

4.Model Selection and Model Training

Model Selection

- Tested multiple regression models to find best accuracy
- Random Forest — Baseline with stable performance
- XG Boost — Lowest error, best accuracy (**Final Model**)
- Light GBM — Fast and efficient with similar results
- **XG Boost selected for final predictions**



Model Training

- Split data into train and test sets
- Trained models on engineered features
- Tuned hyperparameters for best performance

5.Model Evaluation

- Used regression metrics
- MAE → Average error
- RMSE → Penalizes large errors
- R^2 Score → Variance explained

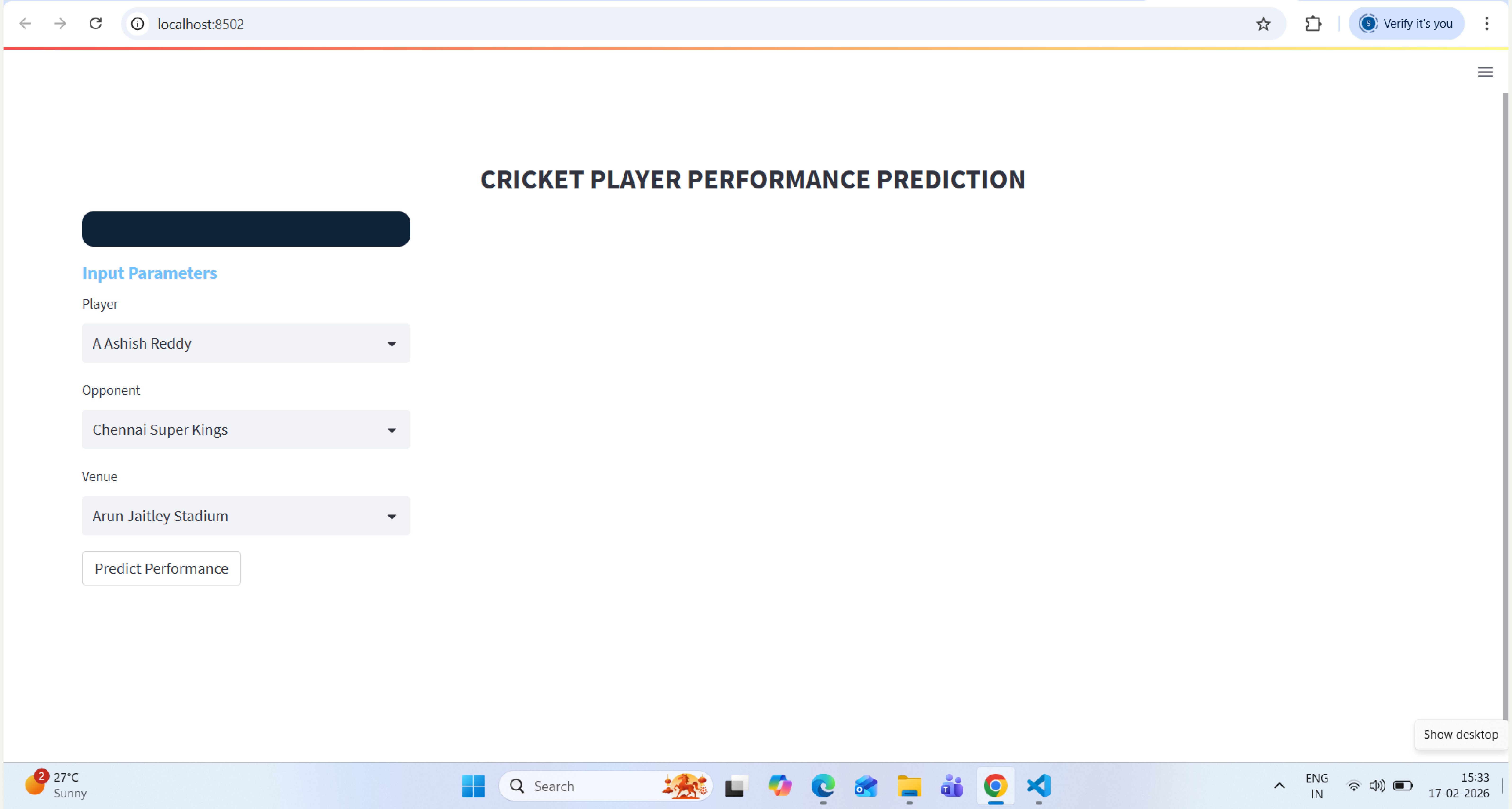
6.Dashboard Development

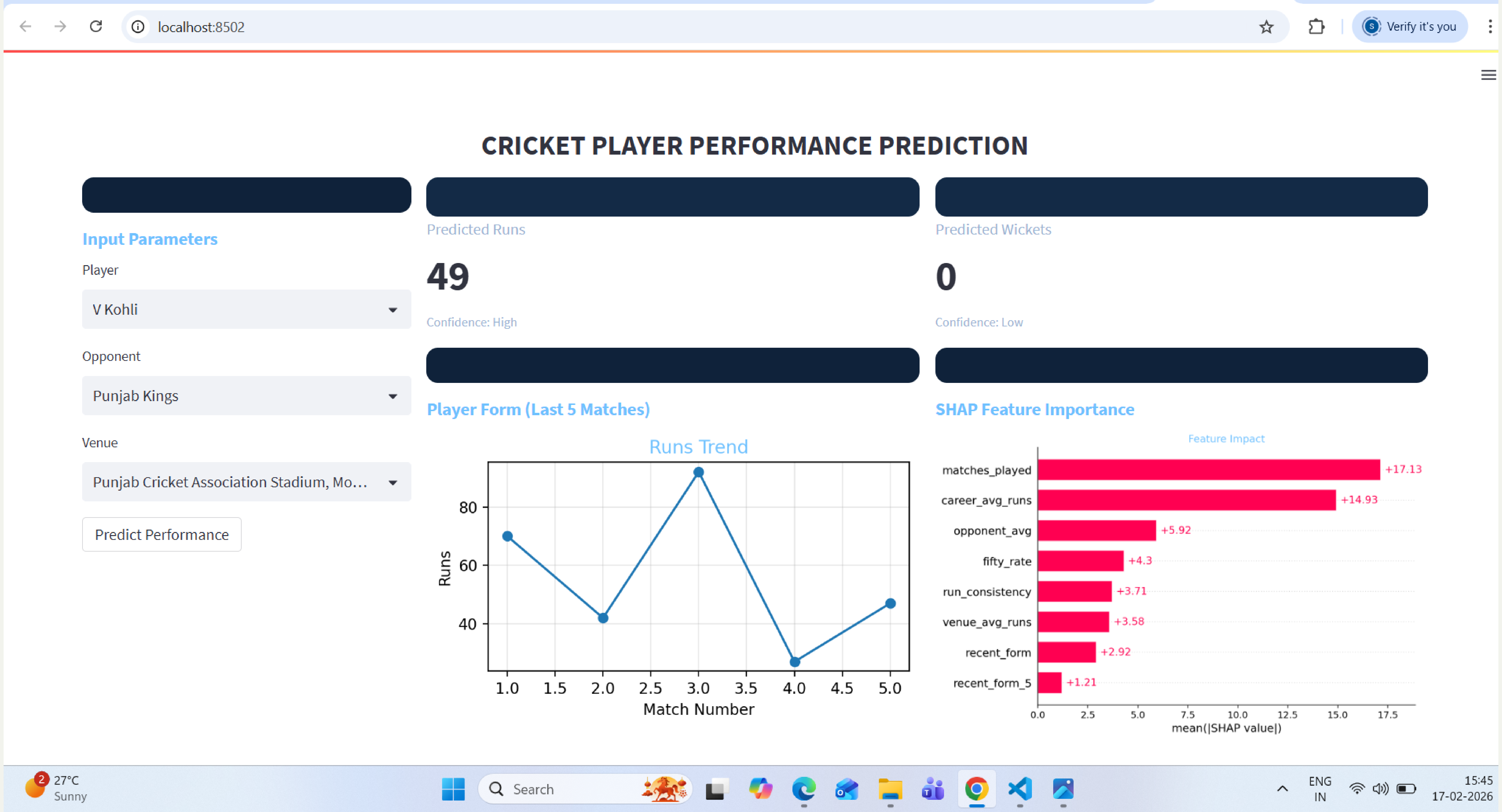
- Built interactive app using Streamlit

Users can:

- Select player and conditions
- View predicted runs & wickets
- Analyze recent form
- Interpret feature importance

Results


localhost:8502
Verify it's you
CRICKET PLAYER PERFORMANCE PREDICTION
Input Parameters
Player
A Ashish Reddy
Opponent
Chennai Super Kings
Venue
Arun Jaitley Stadium
Predict Performance
Show desktop
27°C
Sunny
Search
ENG
IN
15:33
17-02-2026


localhost:8502
Verify it's you
CRICKET PLAYER PERFORMANCE PREDICTION
Input Parameters
Player
V Kohli
Opponent
Punjab Kings
Venue
Punjab Cricket Association Stadium, Mo...
Predict Performance
Predicted Runs
49
Confidence: High
Predicted Wickets
0
Confidence: Low
Player Form (Last 5 Matches)
Runs Trend
Runs
Match Number
SHAP Feature Importance
Feature Impact
matches_played +17.13
career_avg_runs +14.93
opponent_avg +5.92
fifty_rate +4.3
run_consistency +3.71
venue_avg_runs +3.58
recent_form +2.92
recent_form_5 +1.21
mean(|SHAP value|)
27°C
Sunny
Search
ENG
IN
15:45
17-02-2026

Conclusion

The project successfully demonstrates the application of machine learning in sports analytics by predicting cricket player performance using historical data and contextual features. The system not only provides accurate predictions but also delivers interpretable insights through visualizations and feature importance analysis. This work highlights the potential of AI-driven tools in enhancing strategic decision-making in cricket and sets the foundation for more advanced predictive analytics systems.

Thank You